BCMI
Brain-like Computing
&
Machine Intelligence

Generalization and Model Selection

School of Computer Science
Shanghai Jiao Tong University

Outline

- Statistical decision theory
- The Bias-Variance Decomposition
- In-Sample Prediction Error
- Model Selection Criteria
  - AIC, BIC, MDL
- Vapnik-Chernovenkis Dimension
- Cross-Validation

Statistical decision theory

Case 1: Quantitative output Y

Loss function $L(Y, f(X))$ for penalizing errors in prediction.

Most common and convenient loss is ***squared error loss***:

$$L\left(Y, f(X)\right) = (Y - f(X))^2$$

This leads us to a criterion for choosing f.

$$EPE(f) = E(Y - f(X))^2$$

Minimizin $EPE(f)$ leads to a solution:

$$f(x) = E(Y|X = x)$$

Regression Function

$$
\begin{aligned}
EPE(f) &= E[Y - f(X)]^2 \\
&= \int (y - f(x))^2 pr(dx, dy) \\
&= \int (y - f(x))^2 pr(dy \mid dx) pr(dx) \\
&= E_X E_{Y|X} \left([Y - f(X)]^2 \mid X \right)
\end{aligned}
$$

- 对EPE逐点极小化得： $f(x) = \arg\min_c E_{Y|X} \left([Y - c]^2 \mid X = x \right)$
- 极小解为：

$$f(x) = E(Y \mid X = x)$$

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Bias-Variance Decomposition

- Basic Model: $Y = f(X) + \varepsilon, \quad \varepsilon \sim N(0, \sigma_\varepsilon^2)$

- The expected prediction error of a regression fit $\hat{f}(X)$

$$
\begin{aligned}
Err(x_0) &= E[(Y - \hat{f}(x_0))^2 \mid X = x_0] \\
&= \sigma_\varepsilon^2 + [E\hat{f}(x_0) - f(x_0)]^2 + E[E\hat{f}(x_0) - \hat{f}(x_0)]^2 \\
&= \sigma_\varepsilon^2 + Bias(\hat{f}(x_0))^2 + Var(\hat{f}(x_0)) \\
&= \text{Irreducible Error} + \text{Bias}^2 + \text{Variance}
\end{aligned}
$$

- The more complex the model, the lower the (squared) bias but the higher the variance.

Bias-Variance Decomposition

- For the k-NN regression fit the prediction error:

$$Err(x_0) = E[(Y - \hat{f}(x_0))^2 | X = x_0]$$
$$= \sigma_\varepsilon^2 + [f(x_0) - \frac{1}{k}\sum_{j=1}^{k} f(x_j)]^2 + \sigma_\varepsilon^2 / k$$

$$\hat{f}(x_0) = \frac{1}{k}\sum_{x_j \in N(x_0)} f(x_j),$$

- The in-sample error of the Linear Model

$$\frac{1}{N}\sum_{i=1}^{N} Err(x_i) = \sigma_\varepsilon^2 + \frac{1}{N}\sum_{i=1}^{N}[f(x_i) - E\hat{f}(x_i)]^2 + \boxed{\frac{p}{N}\sigma_\varepsilon^2}$$

  – The model complexity is directly related to the number of parameters p.

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Question?

- How to describe the generalization performance of model?
  - Cross-validation
  - Resampling?

In-Sample Prediction Error

- Training Error < True Error

$$\text{Training Error} \quad \overline{err} = \frac{1}{N}\sum_{i=1}^{N} L(y_i, \hat{f}(x_i))$$

$$\text{True Error} \quad Err = E[L(Y, \hat{f}(X)]$$

- Err is extra-sample error
- The in-sample error

$$Err_{in} = \frac{1}{N}\sum_{i=1}^{N} E_{Y^{new}}\left[L(Y_i^{new}, \hat{f}(x_i))|\mathrm{T}\right]$$

- Optimism:

$$\boxed{op \equiv Err_{in} - \overline{err}}$$

In-Sample Prediction Error

- The average optimism is the expectation of the optimism over training sets

$$\boxed{\omega \equiv \mathrm{E}_y(op) \equiv \mathrm{E}_y(Err_{in} - \overline{err})}$$

$$\text{Training Error} \quad \overline{err} = \frac{1}{N}\sum_{i=1}^{N} L(y_i, \hat{f}(x_i))$$

$$\text{In sample error:} \quad Err_{in} = \frac{1}{N}\sum_{i=1}^{N} E_{Y^{new}}\left[L(Y_i^{new}, \hat{f}(x_i))|\mathcal{T}\right]$$

$$\omega = E_y(op) = E_y\left[Err_{in} - \overline{err}\right]$$

$$= \frac{1}{N}\sum_{i=1}^{N} E_{Y^{new}} E_y\left[(Y_i^{new} - \hat{f}(x_i))^2 - (y_i - \hat{f}(x_i))^2\right]$$

$$(Y_i^{new} - \hat{f}(x_i))^2 - (y_i - \hat{f}(x_i))^2$$

$$= (Y_i^{new})^2 - 2Y_i^{new}\hat{f}(x_i) - y_i^2 + 2y_i\hat{f}(x_i)$$

$$\hat{y}_i = \hat{f}(x_i)$$

$$E_{Y^{new}}(Y_i^{new})^2 = E_y y_i^2$$

$$E_{Y^{new}} E_y\left[(Y_i^{new} - \hat{f}(x_i))^2 - (y_i - \hat{f}(x_i))^2\right]$$

$$= 2E[y_i, \hat{y}_i] - 2E\hat{y}_i E y_i = \boxed{2Cov(\hat{y}_i, y_i)}$$

In-Sample Prediction Error

- For squared error, 0-1, other loss function：

$$\omega = \mathrm{E}_y(op) = \frac{2}{N}\sum_{i=1}^{N} Cov(\hat{y}_i, y_i)$$

$$Err_{in} = E_y\left(\overline{err}\right) + \frac{2}{N}\sum_{i=1}^{N} Cov(\hat{y}_i, y_i)$$

- $\hat{y}_i$ is obtained by a linear fit with d inputs or basis function, a simplification is:

$$Err_{in} = E_y\left(\overline{err}\right) + 2\frac{d}{N}\sigma_\varepsilon^2, \qquad \sum_{i=1}^{N} \mathrm{cov}(\hat{y}_i, y_i) = d\sigma_\varepsilon^2$$

- If the dimension / the number of Basis Functions increases, Optimism will increase too.
- If the number of training samples, Optimism will decrease

In-sample Prediction Error

- The general form of the in-sample estimates is

$$\hat{E}rr_{in} = E_y[\overline{err}] + \hat{o}p$$

- d parameters are fit under Squared error loss

$$C_p \text{ statistic : } C_p = \overline{err} + 2\frac{d}{N}\hat{\sigma}_\varepsilon^2$$

  – This relationship introduces the Akaike Information Criterion

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Akaike Information Criterion

- Akaike Information Criterion is more generally applicable estimate of Err_{in}

- A set of models $f_\alpha(x)$ with a turning parameter α :

$$\mathrm{AIC}(\alpha) = \overline{err}(\alpha) + 2\frac{d(\alpha)}{N}\hat{\sigma}_\varepsilon^2$$

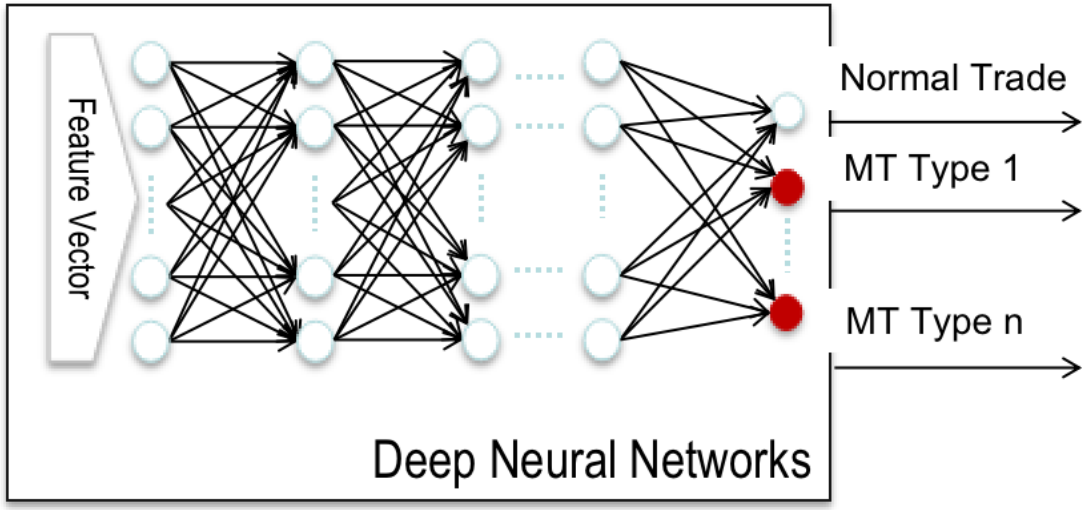
$\overline{err}(\alpha)$:the training error; $d(\alpha)$: number of parameters

- $\mathrm{AIC}(\alpha)$ provides an estimate of the test error curve, and we find the turning parameter $\hat{\alpha}$ that minimizes it.

$$\{f_{\hat{\alpha}}(x) \mid \hat{\alpha} : \min \mathrm{AIC}(\hat{\alpha})\}$$

Model Selection

- To estimate the performance of different models in order to choose the best one.
- What is the parameters for model selection
- CNN – parameter for model selection
  - The number of layers?
  - The number of neurons in each layer?
  - The activation function?
  - The size of convolution kernels?
  - …



Bayesian Approach & BIC

- The Bayesian Information Criterion (BIC)

$$\mathrm{BIC} = -2loglik + (\log N)d$$

- Gaussian model:
  - Variance σ_ε^2

$$-2loglik = C\sum_{i=1}^{N}(y_i - \hat{f}(x_i))^2 / \sigma_\varepsilon^2 = N \cdot \overline{err} / \sigma_\varepsilon^2$$

$$\mathrm{BIC} = \frac{N}{\sigma_\varepsilon^2}[\overline{err} + \boxed{(\log N)\frac{d}{N}\sigma_\varepsilon^2}]$$

  - BIC is proportional to $\mathrm{AIC}(\mathrm{C_p})$, 2 replaced by $\log N$
  - $N > e^2 \approx 7.4,$ BIC tends to choose simple model

Minimum Description Length(MDL)

- Problem：Optimal Coding
  - Message: a b c d
  - Coding： 0 10 110 111
  - Coding2： 110 10 111 0
- **Criterion：** Using the shortest coding length for the most frequent message.
- The probability of zi ： $\Pr(z_i)$
- The coding length ： $l_i = -\log \Pr(z_i)$

Minimum Description Length(MDL)

The Expected Description Lendth

$$E(\text{Length}) \geq -\sum \Pr(z_i)\log \Pr(z_i)$$

$$\text{The equality holds if and only if } p_i = A^{-l_i}。$$

$$Example: \quad \Pr(z_i) = 1/2;\ \ 1/4;\ \ 1/8;\ \ 1/8$$

Minimum Description Length(MDL)

$Model$: M; Parameter: θ; Input Output $Z=(X,y)$

$Suppose$ the Conditional Probabilty function of y

$$p(y\mid\theta,M,X)$$

$$\text{length}=-\log\Pr(y\mid\theta,M,X)-\log\Pr(\theta\mid M)$$

$-\log\Pr(y\mid\theta,M,X)$: the average code length for transmitting the discrepancy between the model and actual target values

$-\log\Pr(\theta\mid M)$: the average code length for transmitting the model parameters

Minimum Description Length(MDL)

$Model$: M; Parameter: θ; Input Output $Z = (X, y)$

$p(y \mid \theta, M, X)$: the Conditional Probabilty function of y,

$p(\theta \mid M)$: the model prior.

MDL Principle: Select a model, minimizing

$$\text{length} = -log\, Pr(\, y \mid \theta, M, x\,) - log\, Pr(\, \theta \mid M\,).$$

$where$ model M is described by the parameter θ

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Vapnik-Chernovenkis Dimension

- Problem：How to select the model dimension d which describes the model complexity
- VC dimension is a critical index describing the model complexity

$$\text{Function family:} \quad \{f(x,\alpha)\}, x \in IR^p$$

$$\alpha -- parameter, f -- index \text{ function}$$

$$Example: \ \alpha = (\alpha_0, \ \alpha_1), \ \text{linear function family}$$

$$f = I(\alpha_0 + \alpha^T{}_1 x > 0); \qquad f \text{ complexity: } \mathrm{p}+1$$

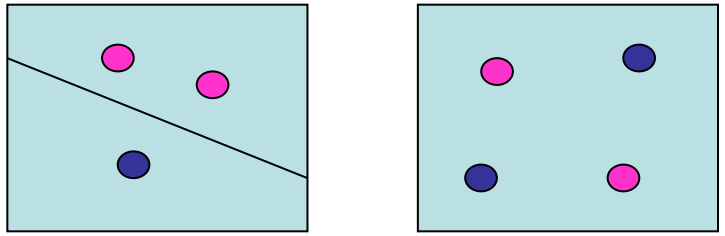
Another example $x \in IR, f(x,\alpha) = I(\sin\alpha \cdot x)$?

f has only one parameter。

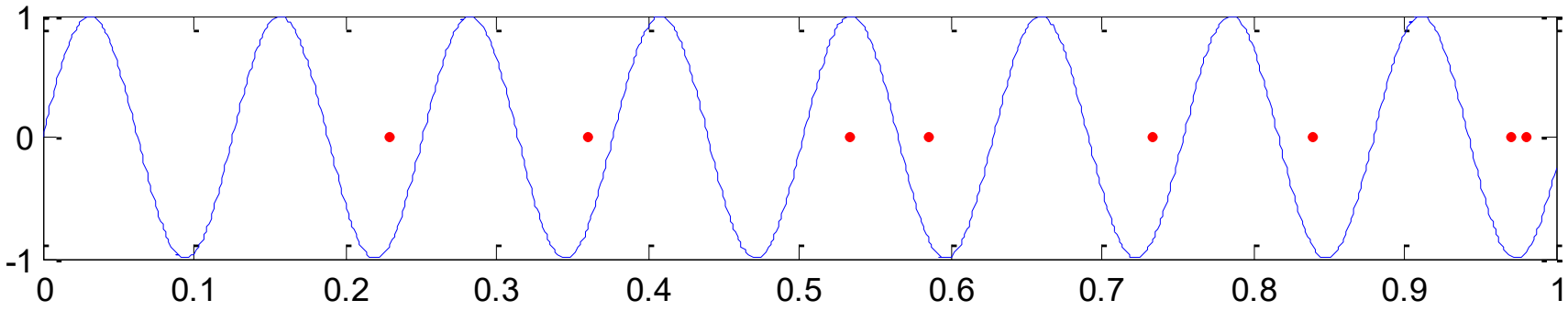
VC dimension

- The VC dimension of function family $\{f(x,\alpha)\}$ is defined to be the largest number *of points that can be* shattered *by members of*

$$\{f(x,\alpha)\}$$



- The VC dimension of 2d linear function family is 3.
- $\{\sin(\alpha x)\}$ its VC dimension is infinite.



VC Dimension

- The VC dimension of a class of real-valued functions $\{g(x,\alpha)\}$ is defined to be the VC dimension of the indicator class $\{I(g(x,\alpha)-\beta>0)\}$ where β takes values over the range of g.

- Assume has VC dimension h, the sample number *N*.

$$Err \le \overline{err} + \frac{\varepsilon}{2}\left(1+\sqrt{1+\frac{4\overline{err}}{\varepsilon}}\right) \quad \text{二类分类}$$

$$Err \le \frac{\overline{err}}{(1-c\sqrt{\varepsilon})_+}; \text{回归} \quad \varepsilon = a_1 \frac{h[\log(a_2 N/h+1)-\log(\eta/4)}{N}$$

$$0 < a_1 \le 4,\ 0 < a_2 \le 2$$

Cherkassky and Mulier (2007, pages 116–118)

Cross Validation

1	2	3	4	5
Training	Training	Testing Set	Training	Training

Cross Validation

- Denote the fitted function by $\hat{f}^{-k}(x)$ with removing k-th fold data. Then the cross-validation estimate of prediction error is

$$\mathrm{CV}(\hat{f}) = \frac{1}{N}\sum_{i=1}^{N} L(y_i, \hat{f}^{-k}(x_i))$$

- The case K = N, *leave-one-out* cross-validation.

- Given model family $f(x,\alpha)$ indexed by a tuning parameter α.
- $\hat{f}^{-k}(x,\alpha)$ ： fited with the kth part of the data removed.

$$\mathrm{CV}(\hat{f},\alpha) = \frac{1}{N}\sum_{i=1}^{N} L(y_i, \hat{f}^{-k}(x_i,\alpha))$$

- Here it looks like a model with about p = 9 predictors would be chosen, while the true model uses p = 10.

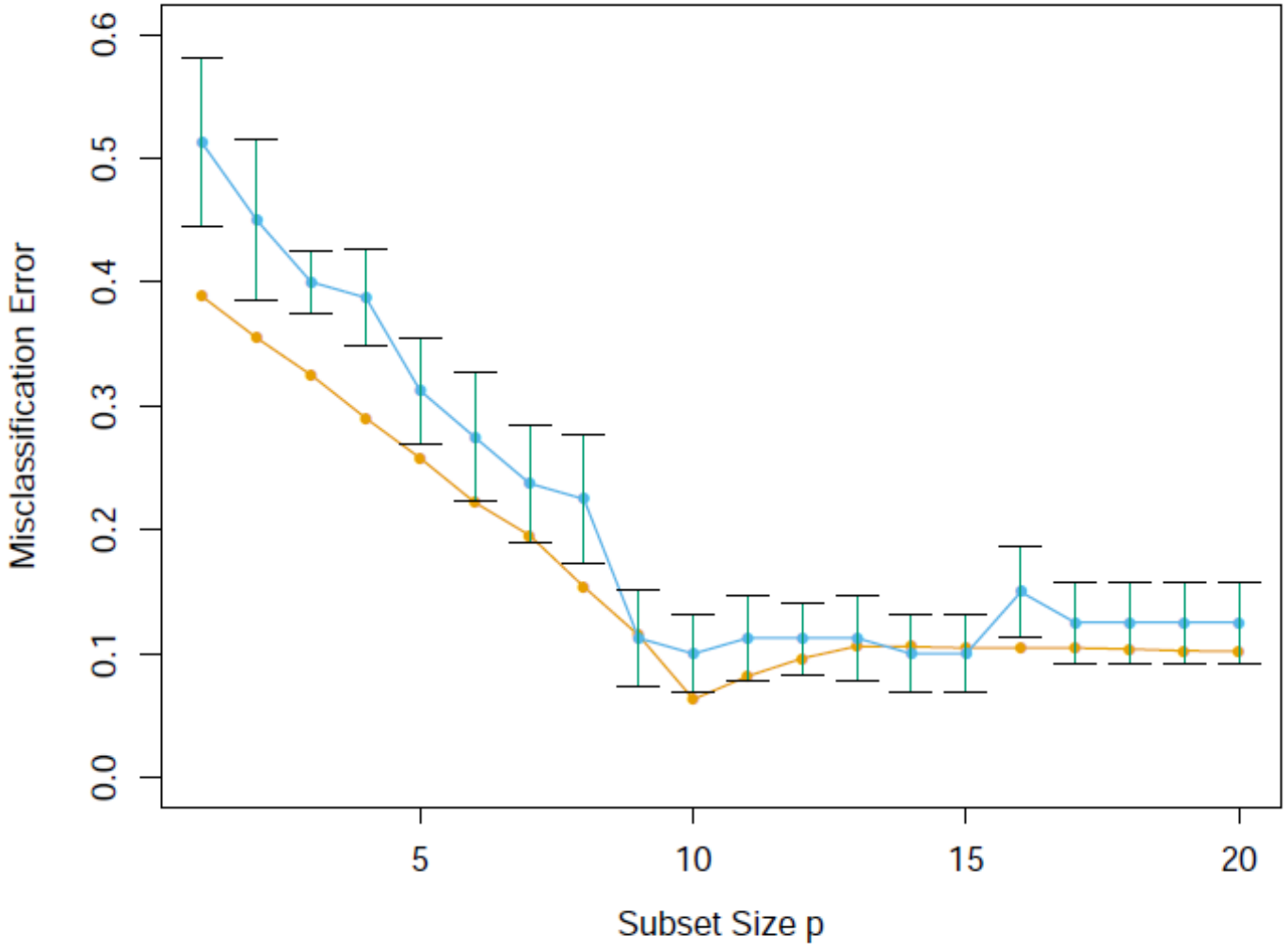


FIGURE 7.9. *Prediction error (orange) and tenfold cross-validation curve (blue) estimated from a single training set, from the scenario in the bottom right panel of Figure 7.3.*

Multiple linear regression

- Statistical Model $y = \beta_0 + \mathbf{x}^T \beta + \varepsilon$
- Model is $y_i = \beta_0 + \mathbf{x}_i^T \beta + \varepsilon_i, \quad i = 1, \cdots, \mathrm{N}$

Equivalently in matrix notation:

$$\mathbf{y} = \mathbf{X}\beta + \boldsymbol{\varepsilon}$$

- $\mathbf{y}$ is N-vector of predicted values
- $\mathbf{X}$ is N $\times$ p matrix of regresses, with 1 in the first column
- β is a p-vector of parameters

Estimation by least squares

- The Cost Function

$$\hat{\beta} = \arg\min \sum_i (y_i - \beta_0 - \sum_{j=1}^{p-1} x_{ij}\beta_j)^2 + \lambda\beta^T\beta$$

$$= \arg\min(\mathbf{y} - \mathbf{X}\beta)^T(\mathbf{y} - \mathbf{X}\beta) + \lambda\beta^T\beta$$

Solution is $\hat{\beta} = (\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T\mathbf{y}$

$$\hat{y} = \mathbf{X}\hat{\beta} = \mathbf{X}(\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T\mathbf{y} \quad \square \quad \mathbf{S}_\lambda\mathbf{y}$$

Generalized cross-validation

- Linear fitting method：

$$\hat{y} = Sy$$

- For linear fitting methods,

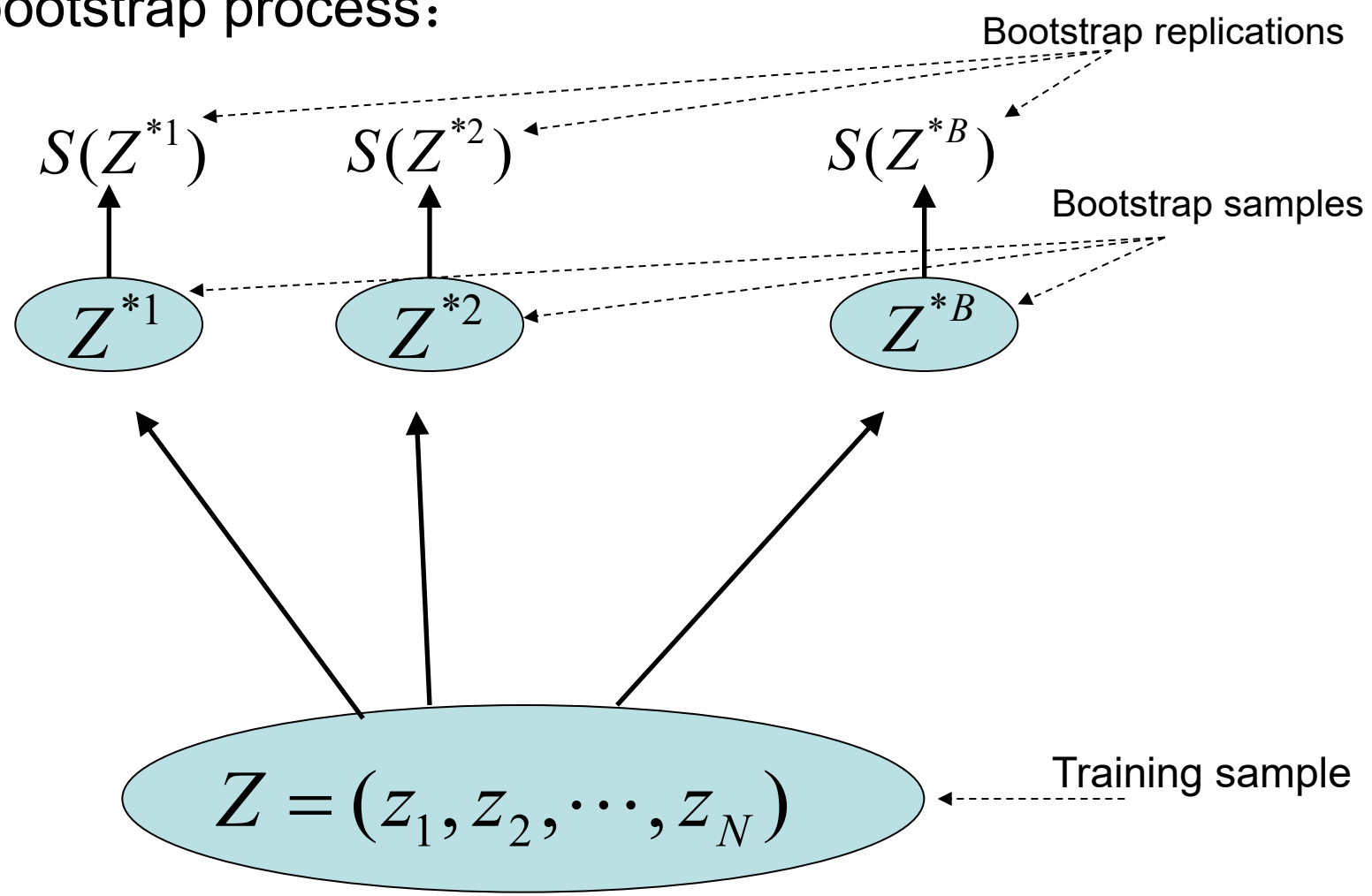
$$\mathrm{CV}(\hat{f}) = \frac{1}{N}\sum_{i=1}^{N}\left[y_i - \hat{f}^{-k}(x_i)\right]^2 = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{y_i - \hat{f}(x_i)}{1 - S_{ii}}\right]^2,$$

- where Sii is the ith diagonal element of S

$$\mathrm{GCV}(\hat{f}) = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{y_i - \hat{f}(x_i)}{1 - trace(S)/N}\right]^2.$$

Bootstrap Methods

- Schematic of the bootstrap process：



Bootstrap Methods

- Basic Idea：The basic idea is to randomly draw datasets with replacement from the training data, each sample the same size as the original training set。
- B times (B = 100 say), producing B bootstrap datasets

$$\hat{f}^{*b}(x_i) \text{ is the prediction function at } x_i$$

- The bootstrap error

$$Err_{boot} = \frac{1}{B}\frac{1}{N}\sum_{b=1}^{B}\sum_{i=1}^{N} L(y_i, \hat{f}^{*b}(x_i))$$

Review of the Talk

- How to derive model selection criterion
  - In-sample error
- What are the most popular model selection criteria
  - AIC; BIC; MDL; VC
- CV for model selection

- Bootstrap method

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The End

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